data can be explained perfectly by imagining that each photon takes both left and right routes toward the detector.

Prochoice

Although we have described the merging of possible histories in the context of only a couple of specific examples, this way of thinking about quantum mechanics is general. Whereas classical physics describes the present as having a unique past, the probability waves of quantum mechanics enlarge the arena of history: in Feynman's formulation, the observed present represents an amalgam—a particular kind of *average*—of all possible pasts compatible with what we now see.

In the case of the double-slit and beam-splitter experiments, there are two ways for an electron or photon to get from the source to the detector screen—going left or going right—and only by combining the possible histories do we get an explanation for what we observe. If the barrier had three slits, we'd have to take account of three kinds of histories; with 300 slits, we'd need to include the contributions of the whole slew of resulting possible histories. Taking this to the limit, if we imagine cutting an enormous number of slits—so many, in fact, that the barrier effectively disappears—quantum physics says that each electron would then traverse *every* possible path on its way to a particular point on the screen, and only by combining the probabilities associated with each such history could we explain the resulting data. That may sound strange. (It is strange.) But this bizarre treatment of times past explains the data of Figure 4.4, Figure 7.1b, and every other experiment dealing with the microworld.

You might wonder how literally you should take the sum over histories description. Does an electron that strikes the detector screen *really* get there by traveling along all possible routes, or is Feynman's prescription merely a clever mathematical contrivance that gets the right answer? This is among the key questions for assessing the true nature of quantum reality, so I wish I could give you a definitive answer. But I can't. Physicists often find it extremely useful to envision a vast assemblage of combining histories; I use this picture in my own research so frequently that it certainly feels real. But that's not the same thing as